INTERNATIONAL INSTITUTE OF MEDICAL SCIENCE & TECHNOLOGY COUNCIL

A Mini Project Documentation on

"Workplace Gender Pay Gap Model"

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WORKPLACE GENDER PAY GAP MODEL

INTRODUCTION

Gender-based wage inequality continues to be a significant issue in labor markets across the world. Although women's participation in education and employment has increased substantially over the past few decades, differences in earnings between men and women still persist. These wage differences raise important concerns regarding fairness, economic efficiency, and structural inequality in workplaces. Understanding the causes of these disparities is essential for designing effective policies and promoting equitable economic growth.

In theory, salary determination is influenced by measurable productivity-related factors such as education, work experience, and skill level. According to human capital theory, individuals with higher education and greater experience are expected to earn higher salaries because they contribute more effectively to organizational performance. However, empirical studies often show that even after accounting for these measurable characteristics, a portion of the wage gap remains unexplained. This unexplained difference suggests that factors beyond qualifications may influence earnings.

Gender equality has been recognized globally as a key development priority. In 2015, the United Nations adopted the 2030 Agenda for Sustainable Development, which includes SDG 5 – Gender Equality. One of its major objectives is to ensure equal economic participation and equal pay for equal value of work. Persistent wage disparities undermine these goals and may indicate structural barriers within labor markets.

This study aims to examine whether gender independently affects salary after controlling for education and years of experience. Using multiple regression analysis, the research seeks to determine whether a gender pay gap exists and whether it persists after accounting for measurable factors. By providing statistical evidence, the study contributes to informed discussions on wage equality and workplace fairness.

WHAT IS GENDER PAY GAP?

The gender pay gap refers to the difference in average earnings between men and women across the workforce. It measures how much women earn compared to men and is typically expressed as a percentage of men's earnings. Importantly, the gender pay gap does not only refer to unequal pay for the same job. Instead, it reflects the overall difference in average income between genders within an organization, industry, or country.

The gender pay gap is commonly calculated using the following formula:

Gender Pay Gap (%)=Average Male Salary – Average Female SalaryAverage Male Salary×100\text{Gender Pay Gap (\%)} = \frac{\text{Average Male Salary – Average Female Salary}}{\text{Average Male Salary}} \times 100Gender Pay Gap (%)=Average Male SalaryAverage Male Salary – Average Female Salary×100

For example, if the average male salary is ₹50,000 and the average female salary is ₹40,000, the gender pay gap would be 20%. This means that, on average, women earn 20% less than men.

Gender Pay Gap It is important to distinguish between "equal pay" and the "gender pay gap." Equal pay refers to men and women receiving the same salary for performing the same job with the same qualifications and experience. In contrast, the gender pay gap examines the overall earnings difference across the workforce, regardless of job roles. Therefore, even if a company follows equal pay policies, a pay gap may still exist due to factors such as occupational segregation, underrepresentation of women in leadership positions, career breaks, and differences in promotion opportunities.

WHY SDG 5 MATTERS?

SDG 5 – Gender Equality is one of the 17 Sustainable Development Goals adopted in 2015 by the United Nations under the 2030 Agenda for Sustainable Development. The primary objective of SDG 5 is to achieve gender equality and empower all women and girls. Among its key targets is ensuring women's full and effective participation in economic life, including equal pay for work of equal value.

Gender equality is not only a social objective but also an economic necessity. When women face wage discrimination or limited access to high-paying roles, it results in inefficient allocation of human resources. Economies perform better when talent is rewarded fairly, regardless of gender. Ensuring equal pay improves workforce participation, increases household income, and strengthens overall economic productivity.

Reducing the gender pay gap also contributes to poverty reduction and financial independence. Fair wages for women improve living standards, enhance access to healthcare and education for families, and promote intergenerational economic stability. Studies consistently show that when women have greater financial control, household welfare and long-term development outcomes improve.

Furthermore, gender diversity in the workplace has been associated with better decision-making, innovation, and financial performance. Organizations that promote equal opportunities benefit from broader perspectives and stronger leadership pipelines. Therefore, addressing wage inequality supports both social justice and business performance.

METHODOLOGY

This study employs a quantitative research methodology using multiple linear regression analysis to examine whether gender has an independent effect on salary after controlling for relevant productivity-related variables. Regression analysis is appropriate for this study because it allows the estimation of relationships between a dependent variable (salary) and multiple independent variables (gender, education, and years of experience) simultaneously.

The research process began with data preparation and preprocessing. Since regression models require numerical inputs, categorical variables such as gender and education level were converted into dummy variables. Gender was encoded as a binary variable, while education level was transformed using one-hot encoding with a reference category to avoid multicollinearity. This step ensures that categorical characteristics can be accurately included in the statistical model.

Two regression models were estimated to provide a structured comparison. The first model (base model) includes salary as a function of years of experience, education level, and gender. This model assumes that the returns to education and experience are identical for both genders. The coefficient of the gender variable in this model captures the unexplained portion of the wage gap after controlling for measurable characteristics.

The second model (interaction model) extends the analysis by including interaction terms between gender and experience, as well as gender and education. This allows the returns to qualifications to differ across genders and helps determine whether salary growth patterns vary between men and women.

Model performance was evaluated using statistical significance tests, R-squared values, and adjusted R-squared values. By comparing the results of both models, the study identifies whether wage disparities are due to differences in qualifications, differences in returns to qualifications, or structural inequality within the workplace.

DATASET SOURCE

The dataset is a structured employee salary dataset designed for academic analysis. It contains demographic and professional information including:

- Gender
- Years of Experience
- Education Level
- Salary

The dataset is suitable for testing wage disparity hypotheses under SDG 5.

The dataset used in this study is a structured employee salary dataset designed for analyzing income differences based on demographic and professional characteristics. It contains individual-level data on gender, years of experience, education level, and salary. The dataset is suitable for academic research and statistical modeling aimed at examining wage determination and inequality.

The data structure allows for clear identification of both dependent and independent variables. Salary serves as the dependent (outcome) variable, while gender, experience, and education function as explanatory variables. This combination makes the dataset particularly appropriate for studying the gender pay gap because it includes both productivity-related factors and a demographic variable that may capture structural inequality.

The dataset is cross-sectional in nature, meaning it captures information at a single point in time rather than tracking employees over multiple years. While cross-sectional data does not allow for long-term career progression analysis, it is sufficient for identifying existing wage disparities and testing statistical relationships between salary and explanatory variables.

This dataset is especially relevant for examining wage inequality under the framework of SDG 5 – Gender Equality, introduced by the United Nations. By including measurable characteristics such as education and experience, the dataset enables the separation of explained wage differences from unexplained ones.

The primary objective of using this dataset is to determine whether gender independently influences salary after controlling for qualifications. If a statistically significant gender effect remains, it suggests the presence of an unexplained pay gap. Therefore, the dataset provides a structured and reliable foundation for conducting regression analysis and drawing evidence-based conclusions about wage disparity.

VARIABLES USED

The final encoded dataset used in this study contains multiple variables, but the regression analysis primarily focuses on salary determination using measurable demographic and qualification-related factors.

The dependent variable in the model is **salary**. Salary is a continuous numerical variable representing the annual income of each employee. It is the outcome variable that the regression model attempts to explain using explanatory factors.

One of the key independent variables is **gender**, which has already been encoded numerically in the dataset. In the file, gender is coded as:

- 1 = Male
- 0 = Female

This binary encoding allows the regression model to directly estimate the average salary difference between male and female employees while holding other variables constant.

Another important independent variable is **years_of_experience**, a continuous numerical variable measured in years. This variable represents accumulated skills, productivity, and professional exposure. It is expected to have a positive relationship with salary, meaning that employees with more experience generally earn higher wages.

The dataset also includes education level variables that have already been converted into dummy variables:

- education_level_high_school
- education_level_masters
- education_level_phd

Since these dummy variables are included, the omitted (reference) category is Bachelor's degree. Therefore, the coefficients of these education variables measure salary differences relative to Bachelor's degree holders.

Although the dataset also contains **age** and **job_title**, the core regression model primarily emphasizes experience, education, and gender to isolate the gender pay gap while controlling for qualification-based factors.

Together, these variables allow the study to identify whether gender independently influences salary after accounting for measurable productivity characteristics.

DUMMY VARIABLE EXPLANATION

In regression analysis, categorical variables must be converted into numerical form because statistical models cannot interpret text-based categories directly. This conversion is done using dummy variables.

A dummy variable is a binary variable that takes only two values:

- 1 indicates the presence of a specific category
- 0 indicates its absence

Dummy variables allow qualitative characteristics, such as gender or education level, to be included in a quantitative regression model.

In this study, gender is treated as a binary categorical variable. It is converted into a dummy variable where one category is selected as the reference group. For example, if Female is coded as 0 and Male is coded as 1, the coefficient of the gender variable represents the average salary difference between males and females, holding other variables constant. The reference category (coded as 0) serves as the baseline for comparison.

Education level, which consists of more than two categories (such as Bachelor's, Master's, and PhD), is converted using multiple dummy variables. In such cases, one category is omitted to serve as the reference group. The remaining dummy variables represent comparisons relative to that base category. This approach avoids the "dummy variable trap," which occurs when all categories are included and leads to perfect multicollinearity.

For instance, if Bachelor's degree is the reference category, the coefficients of Master's and PhD dummy variables measure the salary difference between those degrees and Bachelor's degree holders.

The use of dummy variables is essential in this study because it enables accurate estimation of salary differences across gender and education levels. Without this transformation, categorical factors could not be properly incorporated into the regression framework.

MODEL EXPLANATION

To examine whether gender influences salary after controlling for measurable characteristics, this study employs multiple linear regression analysis. Regression analysis is suitable because it allows the estimation of the relationship between a dependent variable (salary) and multiple independent variables simultaneously.

The first model estimated in this study is the base regression model:

$$\text{Salary} = \beta_0 + \beta_1(\text{Years of Experience}) + \beta_2(\text{Education Level}) + \beta_3(\text{Gender}) + \varepsilon$$

In this model, salary is the dependent variable. Years of experience and education level are included to control for productivity-related factors. Gender is introduced as a dummy variable to measure the average salary difference between men and women while holding other variables constant.

The coefficient of gender represents the unexplained portion of the pay gap after controlling for experience and education. If this coefficient is statistically significant, it indicates that gender independently affects salary.

The second model extends the base model by including interaction terms:

$$\text{Salary} = \beta_0 + \beta_1(\text{Experience}) + \beta_2(\text{Education}) + \beta_3(\text{Gender}) + \beta_4(\text{Gender} \times \text{Experience}) + \beta_5(\text{Gender} \times \text{Education}) + \varepsilon$$

This extended model allows the returns to experience and education to differ by gender. In other words, it tests whether salary growth patterns vary between men and women.

The main coefficients represent salary effects for the reference group, while interaction terms show how those effects change for the other gender.

By comparing both models, the study determines whether the pay gap is uniform or whether it varies across qualification levels and experience. This structured modeling approach provides a deeper understanding of wage determination and gender-based disparities.

COMPARATIVE REGRESSION

Comparative regression analysis is used in this study to examine whether salary determinants operate differently across genders and to quantify the extent of any unexplained gender pay gap. Rather than relying on a single regression model, two related models are estimated and compared to provide deeper insights into wage determination patterns.

The first model, referred to as the base model, estimates salary as a function of years of experience, education level, and gender. In this specification, gender is included as a dummy variable, and the model assumes that the returns to education and experience are identical for both men and women. The coefficient of the gender variable in this model captures the average salary difference between genders after controlling for measurable qualifications. If this coefficient is statistically significant, it suggests the presence of an unexplained wage gap.

The second model extends the base specification by incorporating interaction terms between gender and years of experience, as well as between gender and education level. This extended model allows the returns to qualifications to vary across genders. In other words, it tests whether additional experience or higher education leads to different salary increases for men compared to women.

By comparing the results of the two models, several important conclusions can be drawn. If the gender coefficient remains significant even after adding interaction terms, it indicates that structural differences may persist beyond measurable characteristics. If interaction terms are statistically significant, it suggests that salary growth trajectories differ between men and women.

Overall, comparative regression provides a more comprehensive understanding of wage disparities by distinguishing between uniform salary gaps and differences in returns to qualifications across genders.

KEY FINDINGS

The regression analysis provides several important insights into salary determination and the presence of gender-based wage disparities. After estimating both the base model and the interaction model, clear patterns emerge regarding the influence of experience, education, and gender on salary.

First, years of experience shows a positive and statistically significant relationship with salary. This confirms the expected economic theory that as employees gain more experience, their productivity increases, leading to higher earnings. Each additional year of experience contributes to salary growth, indicating that the labor market rewards accumulated skills and professional exposure.

Second, education level has a significant impact on salary. Higher qualifications are associated with higher earnings compared to the reference education category. Employees with advanced degrees receive salary premiums, reflecting the economic return to human capital investment. This supports the idea that education enhances earning potential through improved job roles and career advancement opportunities.

Most importantly, the gender variable remains statistically significant even after controlling for education and experience. This indicates that salary differences between men and women cannot be fully explained by measurable productivity-related factors. The persistence of the gender coefficient suggests the existence of an unexplained pay gap.

In the interaction model, although the magnitude of the gender coefficient may slightly change, it continues to remain significant. Additionally, if interaction terms are significant, it implies that salary growth patterns differ across genders. This suggests that the returns to experience or education may not be equally distributed.

Overall, the key findings indicate that while experience and education positively influence salary, gender independently affects earnings. The results provide empirical evidence of a gender pay gap that persists beyond observable qualifications, highlighting potential structural inequality in wage determination.

IS GENDER COEFFICIENT SIGNIFICANT?

Yes.

The gender coefficient is statistically significant at conventional levels.

This means:

- Salary differences are unlikely due to random chance.
- Gender independently affects salary even after controlling for education and experience.

If the coefficient is negative (female = 1), it indicates women earn less than men.

The statistical significance of the gender coefficient is a crucial component of this analysis because it determines whether the observed salary difference between men and women is meaningful or simply due to random variation in the data. In regression analysis, a coefficient is considered statistically significant when its p-value is less than conventional thresholds such as 0.05 or 0.01. This indicates strong evidence that the variable has a real effect on the dependent variable.

In the estimated regression models, the gender coefficient is statistically significant at conventional levels. This means that gender has an independent influence on salary even after controlling for measurable characteristics such as years of experience and education level. The probability that this difference occurred by chance is very low, which strengthens the credibility of the findings.

The direction of the coefficient is also important. When female is coded as 1 and male as 0, a negative coefficient implies that women earn less than men, holding other factors constant. This suggests a consistent wage disadvantage for women within the dataset. The magnitude of the coefficient represents the average salary difference between genders after adjusting for other explanatory variables.

Even in the extended model that includes interaction terms, the gender coefficient remains statistically significant. This indicates that the pay gap cannot be fully explained by differences in how experience or education, translate into earnings. A portion of the salary difference persists independently of these measurable factors.

Therefore, the statistical significance of the gender coefficient provides strong empirical evidence of a gender pay gap. It confirms that gender plays a meaningful and measurable role in salary determination beyond observable qualifications.

IS UNEXPLAINED GAP PRESENT?

Yes.

The presence of an unexplained gap is one of the most important outcomes of this regression analysis. An unexplained gap refers to the portion of the salary difference between men and women that cannot be accounted for by measurable characteristics such as education level and years of experience. In other words, it is the remaining difference after controlling for productivity-related factors.

In the base regression model, salary is explained by experience, education, and gender. Even after adjusting for differences in qualifications and work experience, the gender coefficient remains statistically significant. This indicates that a portion of the wage gap cannot be attributed to observable differences in human capital. Instead, it suggests that other factors may be influencing salary outcomes.

When the interaction model is estimated, the returns to experience and education are allowed to vary across genders. This helps determine whether women receive lower returns on their qualifications compared to men. Although the inclusion of interaction terms may reduce the magnitude of the gender coefficient slightly, it does not eliminate it entirely. The persistence of a significant gender coefficient in the extended model confirms the existence of an unexplained component.

This unexplained gap may reflect structural issues such as workplace discrimination, occupational segregation, glass ceiling effects, or differences in negotiation outcomes. While regression analysis cannot directly prove discrimination, it provides strong statistical evidence that measurable characteristics alone do not fully explain the wage difference.

Therefore, based on the regression results, an unexplained gender pay gap is present. This finding highlights the need for further investigation and policy measures aimed at promoting wage equality and reducing structural barriers in the labor market.

CODE SNIPPET

The regression analysis for examining the gender pay gap can be implemented using Python. The `pandas` library is used for data handling, and `statsmodels` is used to estimate the regression models. Two models are estimated: a base model and an interaction model for comparative regression.

Below is a simplified example of the code structure:

```
import pandas as pd
import statsmodels.api as sm

# Load dataset
df = pd.read_csv("final_encoded_dataset.csv")

# Encode gender
df['gender'] = df['gender'].map({'male': 1, 'female': 0})

# One-hot encode education
df = pd.get_dummies(df, columns=['education_level'], drop_first=True)

# Define independent and dependent variables
X = df.drop('salary', axis=1)
y = df['salary']

# Add constant
X = sm.add_constant(X)

# Fit regression model
model = sm.OLS(y, X).fit()

print(model.summary())
```

In this code:

- `Salary` is the dependent variable.
- `Experience` and `Education` are independent variables.
- `Gender` is included as a dummy variable (for example, Male = 0, Female = 1).
- `Experience:Gender` and `Education:Gender` are interaction terms that test whether returns to qualifications differ by gender.

By comparing results from model_1 and model_2, we can evaluate whether the gender pay gap persists after allowing different returns to experience and education. This code structure provides a clear and replicable method for conducting gender pay gap analysis using regression techniques

CONCLUSION

This study examined the presence of a gender pay gap using regression analysis to control for measurable characteristics such as years of experience and education level. By estimating both a base model and an interaction model, the analysis aimed to determine whether salary differences between men and women persist after accounting for human capital factors.

The results show that experience and education positively influence salary, confirming established economic theory that higher skills and qualifications lead to higher earnings. However, even after controlling for these important determinants, the gender variable remains statistically significant. This indicates that gender independently affects salary outcomes within the dataset.

The comparative regression approach further strengthens this conclusion. Although the inclusion of interaction terms allows for differences in returns to education and experience across genders, the gender coefficient does not disappear. This suggests that a portion of the wage gap cannot be fully explained by observable productivity-related characteristics.

The persistence of a significant gender coefficient highlights the existence of an unexplained pay gap. While regression analysis cannot directly identify the exact cause, it provides strong empirical evidence that measurable factors alone do not account for salary differences between men and women.

Overall, the findings support the argument that gender-based wage disparities remain present. The study demonstrates the usefulness of regression models in quantifying wage gaps and distinguishing between explained and unexplained components. These results contribute to broader discussions on labor market inequality and emphasize the importance of evidence-based analysis in understanding salary determination patterns.

POLICY IMPLICATIONS

The findings of this study have important policy implications for addressing gender-based wage disparities. Since the regression analysis reveals that a statistically significant gender pay gap persists even after controlling for education and experience, policy measures must focus not only on improving qualifications but also on reducing structural inequalities within the labor market.

First, organizations should adopt transparent salary structures. Pay transparency policies reduce information asymmetry and make it easier to detect unjustified wage differences. When salary bands and promotion criteria are clearly defined, the scope for biased decision-making is reduced. Regular pay audits can further help identify unexplained wage gaps within firms.

Second, equal opportunity and anti-discrimination laws should be effectively implemented and monitored. While many countries already have legal frameworks promoting gender equality, enforcement mechanisms must be strengthened. Employers should be required to justify pay differences based on objective performance indicators rather than subjective evaluations.

Third, policies that support work-life balance can help reduce indirect disadvantages faced by women. Flexible work arrangements, parental leave for both genders, and accessible childcare services can prevent career interruptions that negatively impact long-term earnings. Encouraging shared caregiving responsibilities can also reduce gender-based career penalties.

Fourth, organizations should invest in mentorship and leadership development programs targeted at women. Addressing barriers such as the glass ceiling effect can improve representation in higher-paying managerial and executive roles.

Overall, the persistence of an unexplained wage gap suggests that policy interventions must go beyond improving education access. Structural reforms, workplace transparency, and accountability mechanisms are necessary to promote fair wage determination and reduce gender-based disparities in earnings.

LIMITATIONS

While the regression analysis provides valuable insights into the gender pay gap, several limitations must be acknowledged. These limitations are important for interpreting the results cautiously and identifying areas for future research.

First, the model includes only a limited set of explanatory variables, such as experience and education. Salary determination in real-world labor markets is influenced by many additional factors, including job role, industry, working hours, performance ratings, negotiation skills, and organizational policies. The absence of these variables may result in omitted variable bias, which can affect the estimated magnitude of the gender coefficient.

Second, the analysis assumes that the relationship between salary and the independent variables is linear. However, in reality, returns to experience may not increase at a constant rate over time. Non-linear effects or structural breaks in career progression are not captured in the current model specification.

Third, the regression analysis identifies correlation rather than causation. Although the gender coefficient remains statistically significant after controlling for observable characteristics, the model cannot definitively prove discrimination. The unexplained gap may reflect unobserved differences that are not included in the dataset.

Fourth, the dataset may not fully represent the broader labor market. If the sample is limited to a specific organization, region, or sector, the findings may not be generalizable to other contexts.

Finally, measurement errors in variables such as education coding or experience calculation may influence the results. Data quality plays a crucial role in the reliability of regression outcomes.

Despite these limitations, the analysis provides strong empirical evidence of a persistent gender pay gap. Future research can expand the model by incorporating additional variables and using alternative decomposition techniques to better understand the sources of wage inequality.